

ADITYA CHUNDURI

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SUMMARY

MS in Applied Data Science from USC with a focus on applied machine learning engineering across LLM systems, audio, and computer vision. Builds reproducible Python pipelines for RAG, local inference, source separation, image segmentation, and model evaluation, served through FastAPI and Gradio with PyTorch, Docker, FAISS, and GitHub Actions. Coauthored a published ML paper, with strength in evaluation, failure mode analysis, and code that other people can run again reliably.

EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Applied Data Science

Jan 2024 to Dec 2025

SRM Institute of Science and Technology

Chennai, India

Bachelor of Technology in Computer Science and Engineering, AI/ML Specialization

Jun 2019 to Jul 2023

PROJECTS

Agent Shield | LLM Agent Security Evaluation Framework · [GitHub](#)

Apr 2026 to Present

Skills: Python, Inspect AI, MCP, AgentDojo, HarmBench

- Built an LLM agent evaluation framework in the Inspect AI harness covering prompt injection, MCP tool misuse, RAG memory poisoning, and behavioral drift across 8 frontier models.
- Defined seeded JSONL tasks and unified scoring (attack success rate, benign utility, transparency rate) so runs stay comparable across models, mapped to OWASP and MITRE ATLAS.

AI RemixMate | AI DJ Mixing Engine · [GitHub](#)

Sep 2025 to Dec 2025

Skills: Python, PyTorch, Demucs, Librosa, FastAPI, Streamlit

- Built a Python DJ engine that renders beat locked, key aware transitions between any two tracks using Demucs stem separation and music information retrieval, with dynamic EQ fades and ITU-R LUFS mastering.
- Served it through a FastAPI backend and Streamlit interface with reusable audio preprocessing modules.

AI Health Journal | Privacy First Local LLM Assistant · [GitHub](#)

Jan 2025 to May 2025

Skills: Python, Flask, Ollama, ChromaDB, RAG, DPO

- Built a privacy first journaling assistant on Flask and Ollama with RAG over ChromaDB, all processing on localhost and no external APIs, plus a DPO fine tuning pipeline to align responses to the user.

Model Behavior Lab | Local LLM Evaluation Platform · [GitHub](#)

Sep 2024 to Dec 2024

Skills: Python, Ollama, JSON test suites, Plotly

- Built a local Ollama based LLM evaluation platform benchmarking reasoning, hallucination, emotion alignment, and code correctness, with repeatable runs and dashboards; the methodology became the base for Agent Shield.

AkashicTree | Automated GenAI Media Pipeline · [GitHub](#)

Jun 2025 to Aug 2025

Skills: Python, Diffusers, Ollama, ElevenLabs, FLUX.1

- Built an agentic GenAI media pipeline that turns a single brief into text, image, and audio, coordinating local Ollama inference, Diffusers and FLUX.1, and ElevenLabs voice through a modular workflow.

EXPERIENCE

USC - Viterbi School of Engineering

Los Angeles, CA

Research Assistant, Computer Vision and Medical Imaging

Aug 2024 to Dec 2024

- Built a PyTorch U-Net segmentation pipeline for retinal artery vein classification on CVD-Masks, using custom loss functions and architecture tuning.
- Designed modular, experiment tracked training pipelines for reproducible model comparison and automated medical image preprocessing to cut manual curation effort.

SSN College of Engineering

Remote

Research Intern, Computer Vision and Object Detection

Jun 2021 to Jul 2021

- Built object tracking pipelines with fine tuned YOLO based CNNs and packaged reusable PyTorch and TensorFlow modules to streamline frame processing experiments.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, Bash

ML and AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Keras, OpenCV, Librosa, Demucs, Diffusers, LangChain

LLM and Eval: RAG, Ollama, ChromaDB, FAISS, Inspect AI, AgentDojo, HarmBench, prompt engineering, adversarial evaluation, benchmarking

Backend and MLOps: FastAPI, Flask, REST APIs, Docker, AWS, Git, GitHub Actions, CI/CD, Gradio, Streamlit

Data and Analysis: PostgreSQL, Pandas, NumPy, Plotly, statistical evaluation, experiment design, model benchmarking

PUBLICATIONS

Wind Power Analysis Using Machine Learning | IJRASET, Aug 2023, Co-Author